

More on Turing Machines and Their Variants

CSCI3360 Introduction to Computational Complexity
Tutorial (Week 2)

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TA information

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Fun fact	I was a CUHK CS undergraduate too! :)

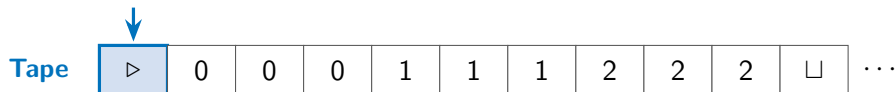
I recommend emailing me before visiting my office, in case I am unexpectedly away. You are also welcome to drop by whenever I am in the office.

Practice Problem Policy

- A practice problem set will be released every one or two tutorials. Each set contains 7–8 questions.
- These problems are ungraded. They are intended to help you understand the course material and prepare for quizzes and exams. (Don't rely solely on them, as they cover only part of the course content. The course note is the primary reference.)
- Solutions will be provided for only half of the problems in each set.

Machine Model

As in Problem Set 1: one tape and one read/write head.



- $\Sigma = \{0, 1, 2\}$. The work symbol $X \notin \Sigma$ marks a processed symbol.
- The head moves L , S , or R . Never overwrite or move left of $\triangleright$.
- Marking preserves positions. The first $\square$ still marks the input's end.

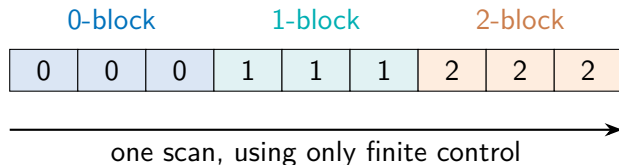
Algorithmic Description

Deciding $L = \{0^k 1^k 2^k : k \geq 0\}$. On input x :

- ① Check $x \in 0^* 1^* 2^*$. Otherwise, output 0 and halt.
- ② Return to the beginning. Find an unmarked 0 and mark it X . If no 0 remains, go to the final check.
- ③ Scan right for an unmarked 1 and mark it X . Reject if a 2 or $\sqcup$ appears first.
- ④ Scan right for an unmarked 2 and mark it X . Reject if $\sqcup$ appears first. Repeat from Step 2.

Final check: Scan the whole input. Output 1 if every symbol is X , and 0 otherwise. Halt.

Step 1: checking the block order

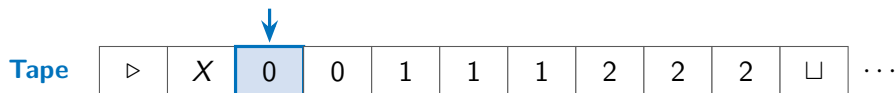


The symbols may advance from 0 to 1 to 2, but never go backwards.

010122 fails when a 0 appears after a 1. Empty blocks are allowed at this stage.

Round 1: marking one zero

Input: 000111222. Mark the first 0 and move right.

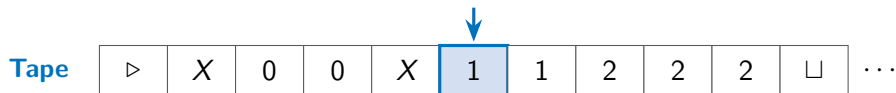


$$0 \mapsto X.$$

The state now records: **searching for a matching 1**. Skip the remaining 0's and any X's.

Round 1: finding and marking one one

Scan right across the remaining 0's, then mark the first 1.

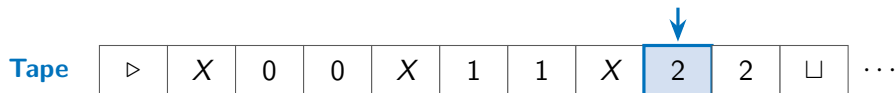


The selected 0 now has a matching 1.

The state changes to **searching for a matching 2**. Skip the remaining 1's and any X's.

Round 1: finding and marking one two

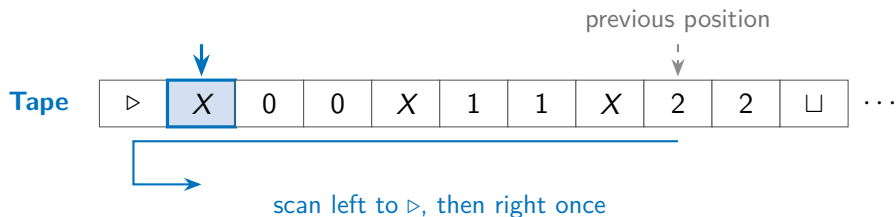
Scan right across the remaining 1's, then mark the first 2.



We have marked exactly one 0, one 1, and one 2.

This completes one round. The next task is to return to the beginning.

Returning to the beginning

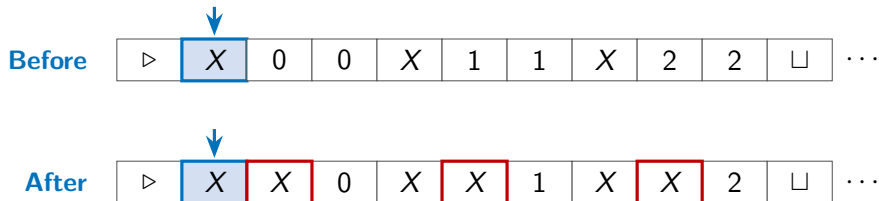


Move left until the head sees $\triangleright$, then right once.

This is a sequence of single-cell moves. The permanent marker stays unchanged.

All X 's remain on the tape for later rounds.

Round 2: one more symbol from each block

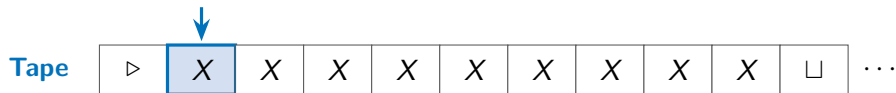


The red outlines show the three new marks in this round.

Two 0's, two 1's, and two 2's have now been processed.

Round 3: the final triple

Mark the last 0, the last 1, and the last 2, then rewind.

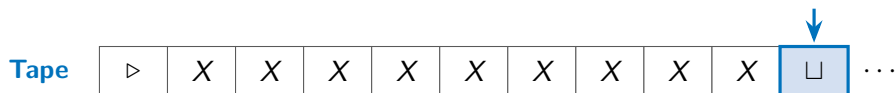


Every original input position now holds X .

The next search finds no unmarked 0, so the machine enters the final check.

Final check: accept

Scan from the beginning to the first blank.



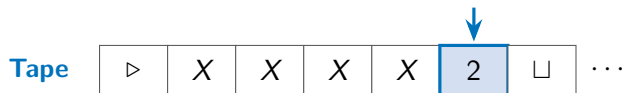
No unmarked 1 or 2 remains.

Output 1 and halt.

This input contains exactly three symbols of each kind, in the required order.

Rejection: a matching one is missing

Input: 00122. After one full round, start the second round.

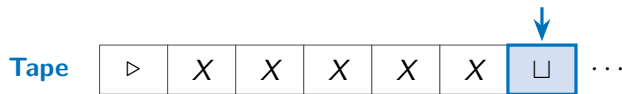


The second 0 is marked, but the search for its 1 reaches a 2.

There is no unmarked 1 left. **Output 0 and halt.**

Rejection: a matching two is missing

Input: 00112. The second round has marked a 0 and a 1.

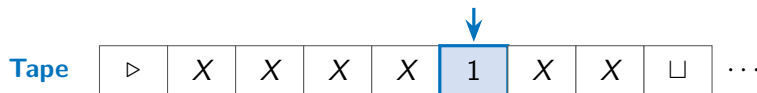


The head reaches □ while **searching for a 2**. Output 0 and halt.

Even though every input symbol is marked, this round is incomplete. The control state matters too.

Rejection: an unmarked symbol remains

Input: 0011122. After two complete rounds:



No unmarked 0 remains, but the final check finds an unmarked 1.

The counts are unequal. **Output 0 and halt.**

An extra 2 would fail the same final check.

Practice: which check decides the answer?

Two minutes with a neighbour. Give the first failed check, or explain acceptance.

A 0122

B 0012

C 010122

D ε

Why does the algorithm use a nonblank mark X ?

Practice: answers

0122 One round completes. An extra 2 remains in the final check.

0012 In round 2, the search for a 1 reaches blank.

010122 The order check sees a 0 after a 1.

ε No rounds are needed. The final check finds no unmarked symbol.

$X \neq \sqcup$ preserves the input positions and the location of the input's end.

Correctness

After the order check, write $x = 0^a 1^b 2^c$.

Invariant. After r completed rounds, exactly r symbols from each block have been marked.

- ① **If $x \in L$:** $a = b = c$. Every round finds both partners. After a rounds, the final check outputs 1.
- ② **If $x \notin L$:** The order check fails, a partner is missing, or an unmarked 1 or 2 remains after all 0's are used. The machine outputs 0.

Halting and Running Time

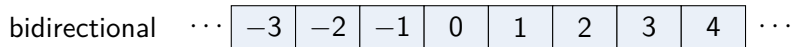
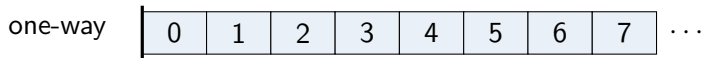
For every input, the machine halts.

- Each completed round marks three new positions. There are at most $\lfloor |x|/3 \rfloor$ completed rounds.
- Every forward scan is bounded by the first $\sqcup$. Every rewind ends at $\triangleright$.

A round uses $O(|x|)$ moves. The total is $O(|x|^2)$.

This is the bound for the **single-tape marking algorithm**.

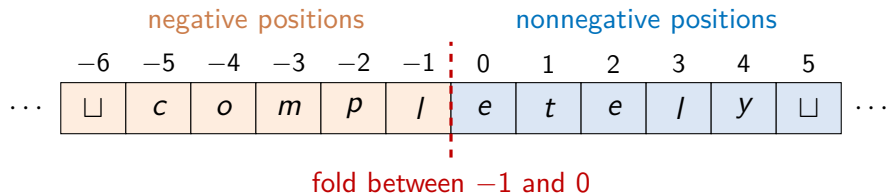
One-way and bidirectional tapes



Both heads can move left and right. “One-way” describes the extent of the tape.

Can one physical tape with a left endpoint represent both halves?

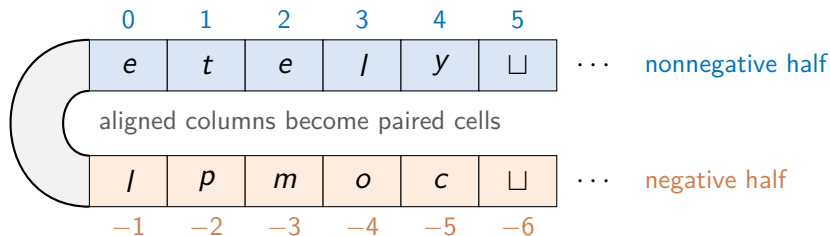
Before the fold: a tape indexed by integers



We choose the fold between positions -1 and 0 .

The nonnegative half starts with *e t e l y*. The negative half contains *c o m p l* in the original order.

The negative half folds underneath



After folding, the lower row runs $-1, -2, -3, \dots$ from left to right. Its symbols appear as *l p m o c*.

Each aligned column pairs a nonnegative cell with a negative cell.

One physical cell stores a pair of symbols

		0	1	2	3	4	5	
P	▷	e	t	e	l	y	□	...
N		l	p	m	o	c	□	

paired data cell j stores (a_j, a_{-j-1}) .

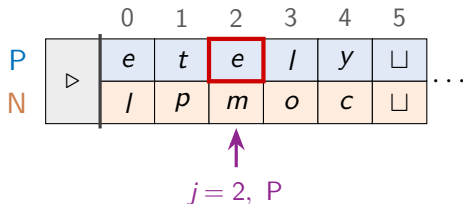
P selects the upper component, and N selects the lower component.

The first cell stores (e, l) , the next (t, p) , and then (e, m) .

Each column is one symbol in $\Gamma \times \Gamma$. A permanent $\triangleright$ precedes paired cell 0.

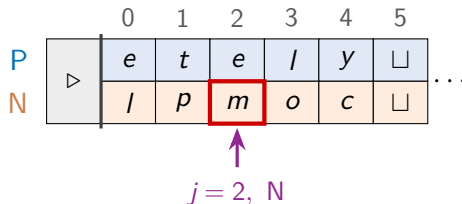
The head position also needs a mode

Simulated position 2



Paired cell 2, mode P.

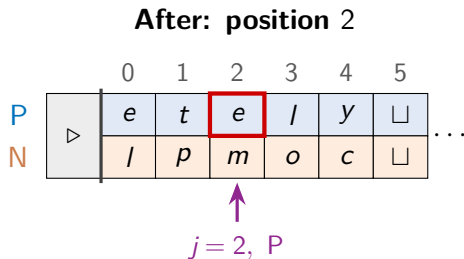
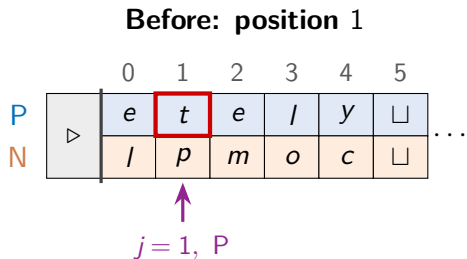
Simulated position -3



Paired cell 2, mode N.

The finite control records P or N to select the active component. The purple arrow marks the single physical head.

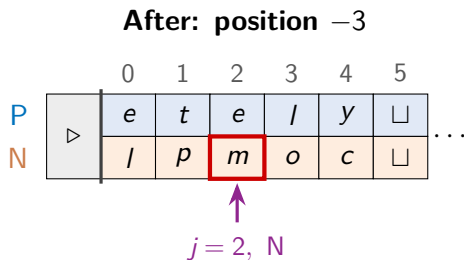
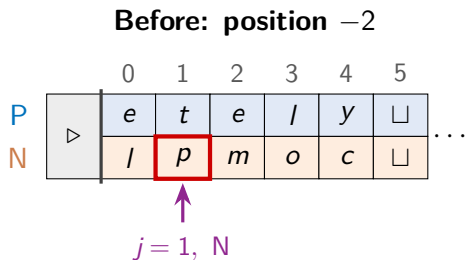
Moving on the nonnegative half



Simulated move: R . **Physical move:** R .

On the nonnegative half, physical and simulated directions agree. The mode stays P .

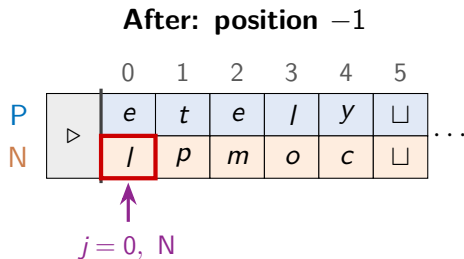
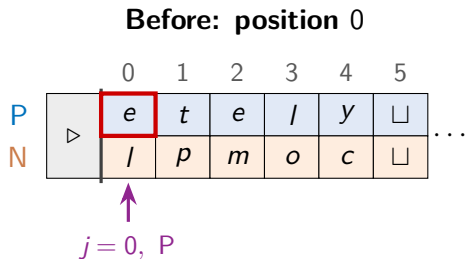
Moving on the negative half



Simulated move: L . **Physical move:** R .

More negative positions lie farther right in the lower row. The directions reverse, and the mode stays N .

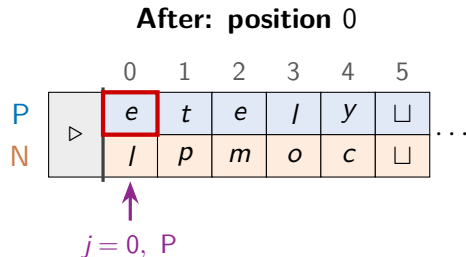
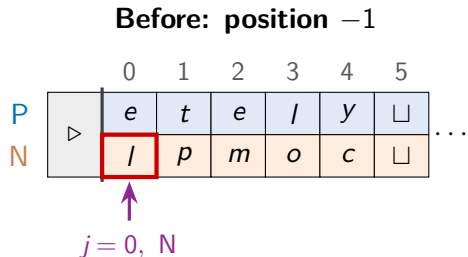
Crossing the fold: 0 to -1



Simulated move: L . **Physical move:** S .

Both symbols are in pair 0. The physical head stays still, and the mode switches from P to N .

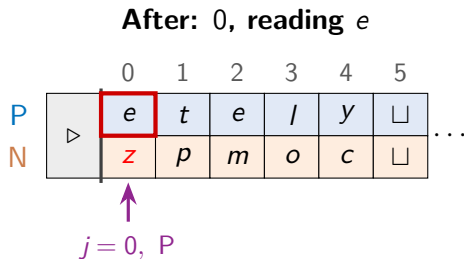
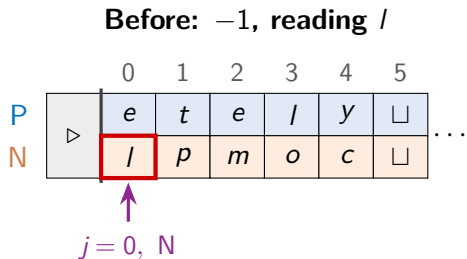
Crossing back: -1 to 0



Simulated move: R . **Physical move:** S .

Switch from N to P in pair 0 . **A crossing changes the active component, not the physical cell.**

Writing a symbol, then crossing the fold



The simulated instruction writes z and moves right from position -1 .

$$(e, /) \mapsto (e, z), \quad N \mapsto P.$$

Update the component selected by the **old mode**, then change the mode.

Movement rules

Current mode	Simulated move	Physical move	Next mode
P	R	R	P
$P, j > 0$	L	L	P
$P, j = 0$	L	S	N
N	L	R	N
$N, j > 0$	R	L	N
$N, j = 0$	R	S	P

A simulated S leaves the physical head and mode unchanged.

At the fold: 0 and -1 share paired cell 0. Crossing between them switches the mode while the head stays still.

Pair exercise: a walk across the fold

The simulated head starts at position 0 and makes these moves:

$L, \quad L, \quad R, \quad R, \quad R.$

Work with a neighbour for 90 seconds. Record the physical cell and mode after every move.

Step	Start	L	L	R	R	R
Simulated position	0	-1	-2	-1	0	1
Paired cell	0	?	?	?	?	?
Mode	P	?	?	?	?	?

Which two moves cross the fold?

The same walk on the one-way tape

Step	Start	L	L	R	R	R
Simulated position	0	-1	-2	-1	0	1
Paired cell	0	0	1	0	0	1
Mode	P	N	N	N	P	P
Physical move	-	S	R	L	S	R

Crossings: The first and fourth moves switch the mode at paired cell 0.

Writing: In mode N, writing b changes (u, v) to (u, b) . The other component stays unchanged.

Why the simulation works

Invariant after every simulated step

- Cell j stores (a_j, a_{-j-1}) .
- The physical head and mode identify the simulated head.
- The finite control also stores the simulated state.

Read and update the active component, then use the movement table. This preserves the invariant in $O(1)$ steps.

Resources: A finite paired alphabet $\Gamma \times \Gamma$, a mode bit, and a marker for the fold suffice for one tape.

T simulated steps cost $O(T)$ after setup. Encoding an input of length n gives total $O(n + T)$, or $O(T)$ when $T \geq n$.

What the two constructions show

Algorithm design on one tape

Each round marks one 0, one 1, and one 2. The invariant proves correctness, and the bound on rounds proves halting.

Simulation by folding

Pair the two halves of the tape. The physical head position and the mode identify the simulated head, including when it crosses the fold. The two tape models compute the same functions.

In your problem-set solutions: Give the algorithmic description, prove both output cases, and explain why every computation halts.